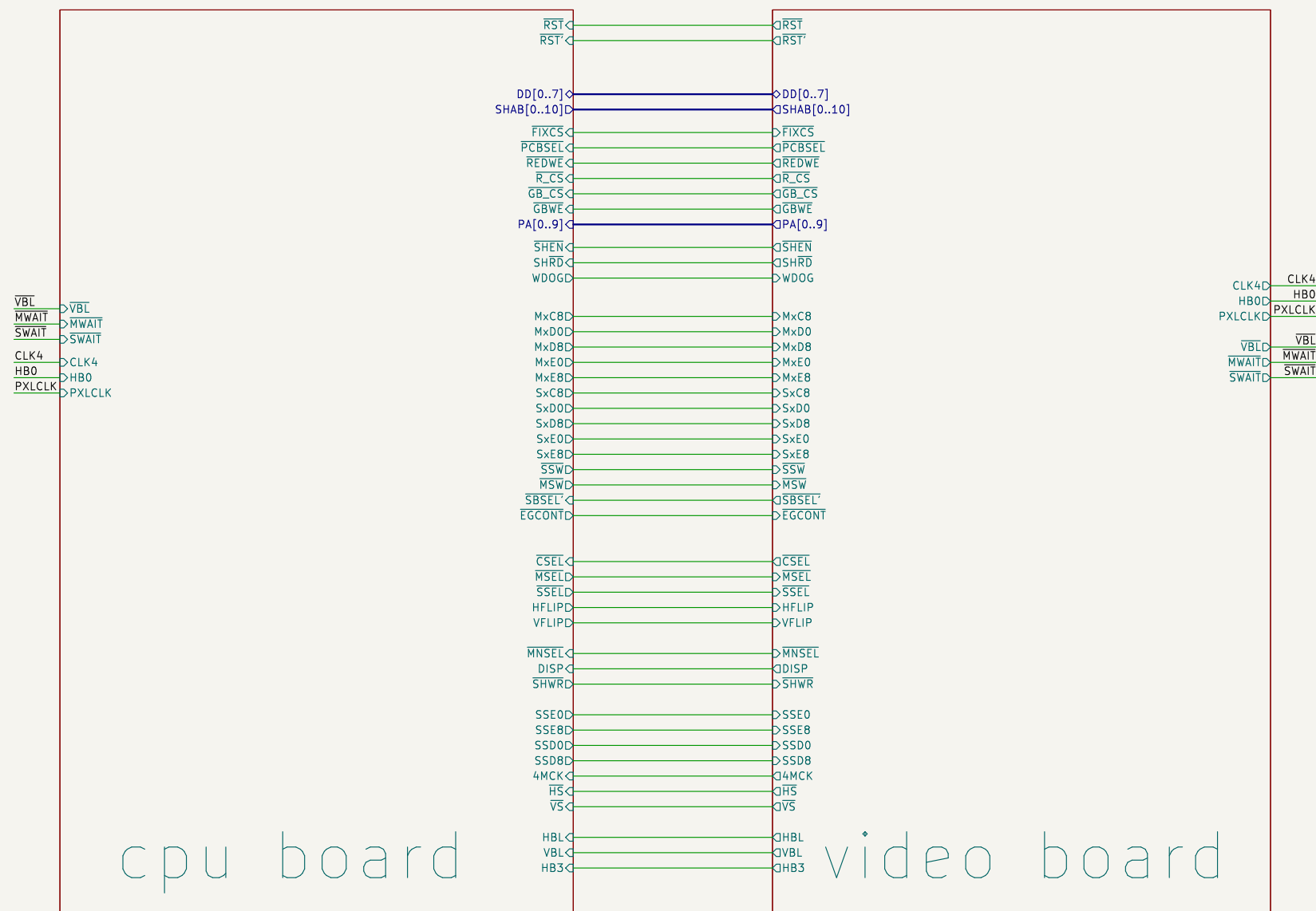




cpu board

video board

```
tehkanwch
bootleg board
```

- original PCB has through-hole vias whereas the bootleg has filled vias
 - original uses three custom packages from Mitsubishi with logic inside.
- These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
 - debug edge connector on video board not present on bootleg
 - cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
 - different power amplifier for sound without heat sink on the bootleg
 - slightly different frequencies for crystal oscillators (cost reduction?)
 - device reference markers (silkscreen) on bootleg have reversed order on bootleg

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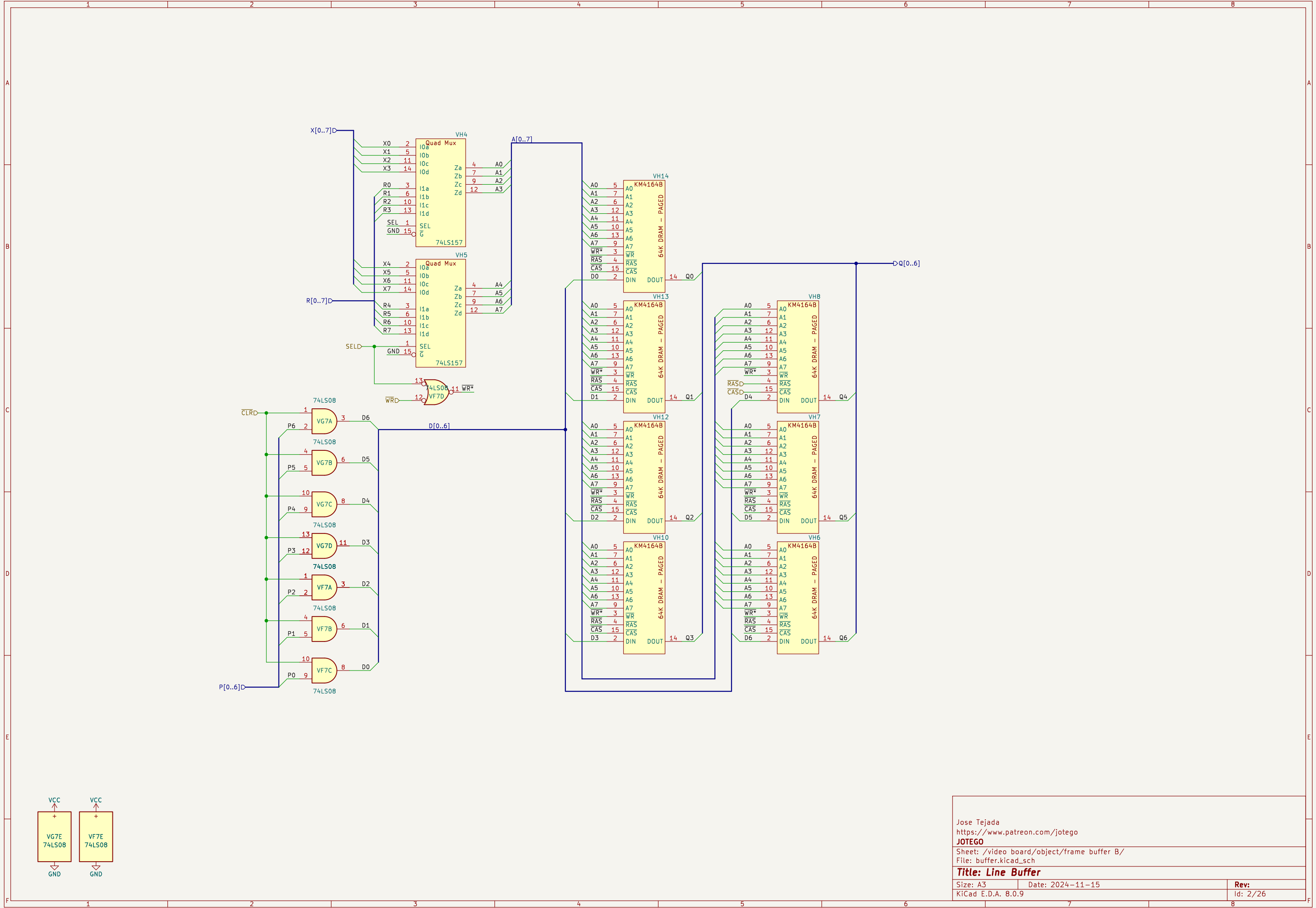


Sheet: /
File: tehkanwch.kicad_sch

Title: TEHKAN WORLD CUP (tehkanwch bootleg)

Size: A3	Date: 2024-11-15
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Rev: 1/26



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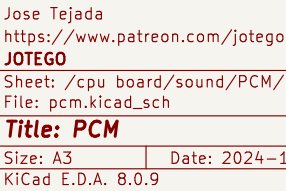
JOTEGO

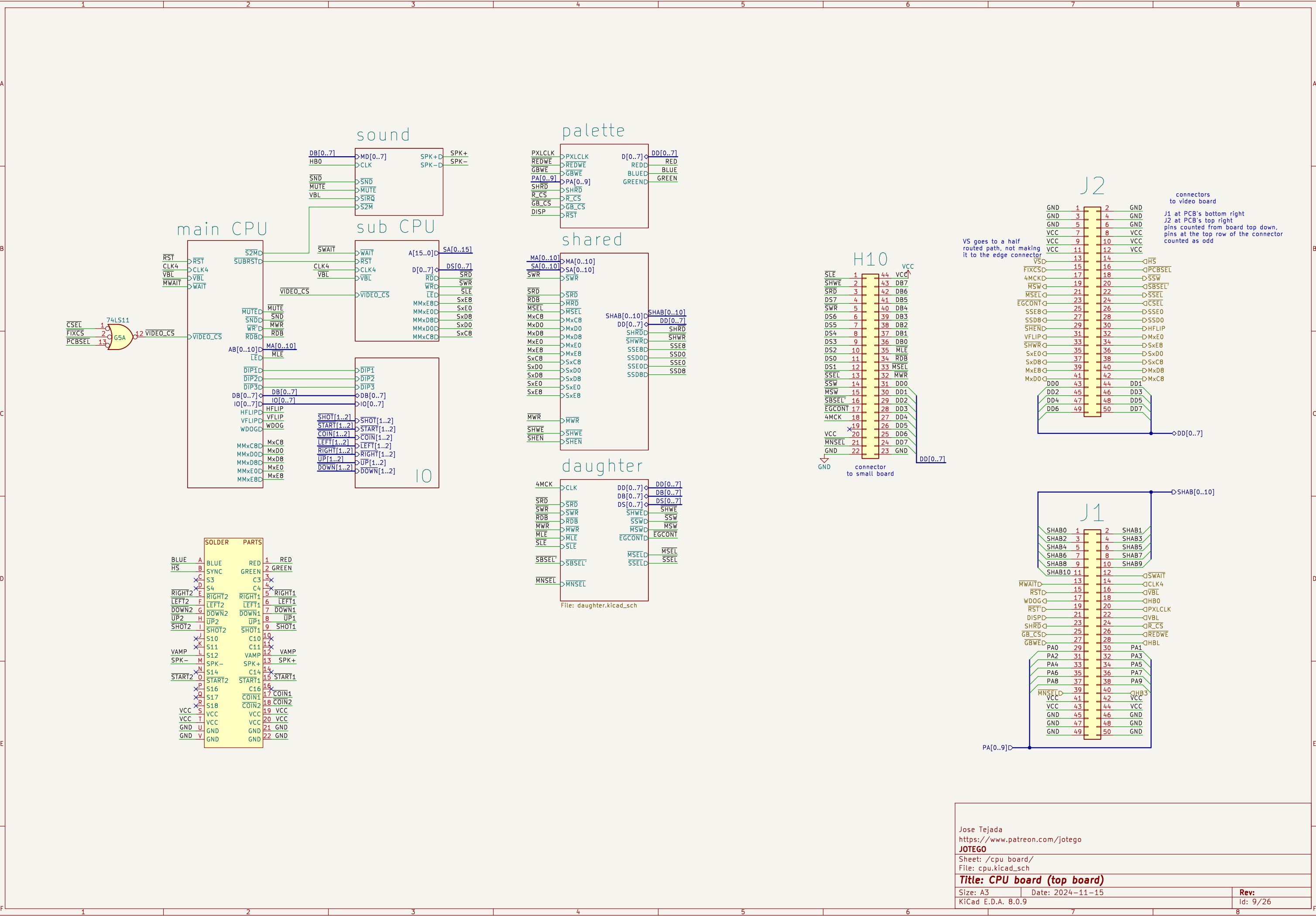
Sheet: /video board/object/frame buffer B/
File: buffer.kicad_sch

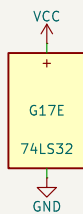
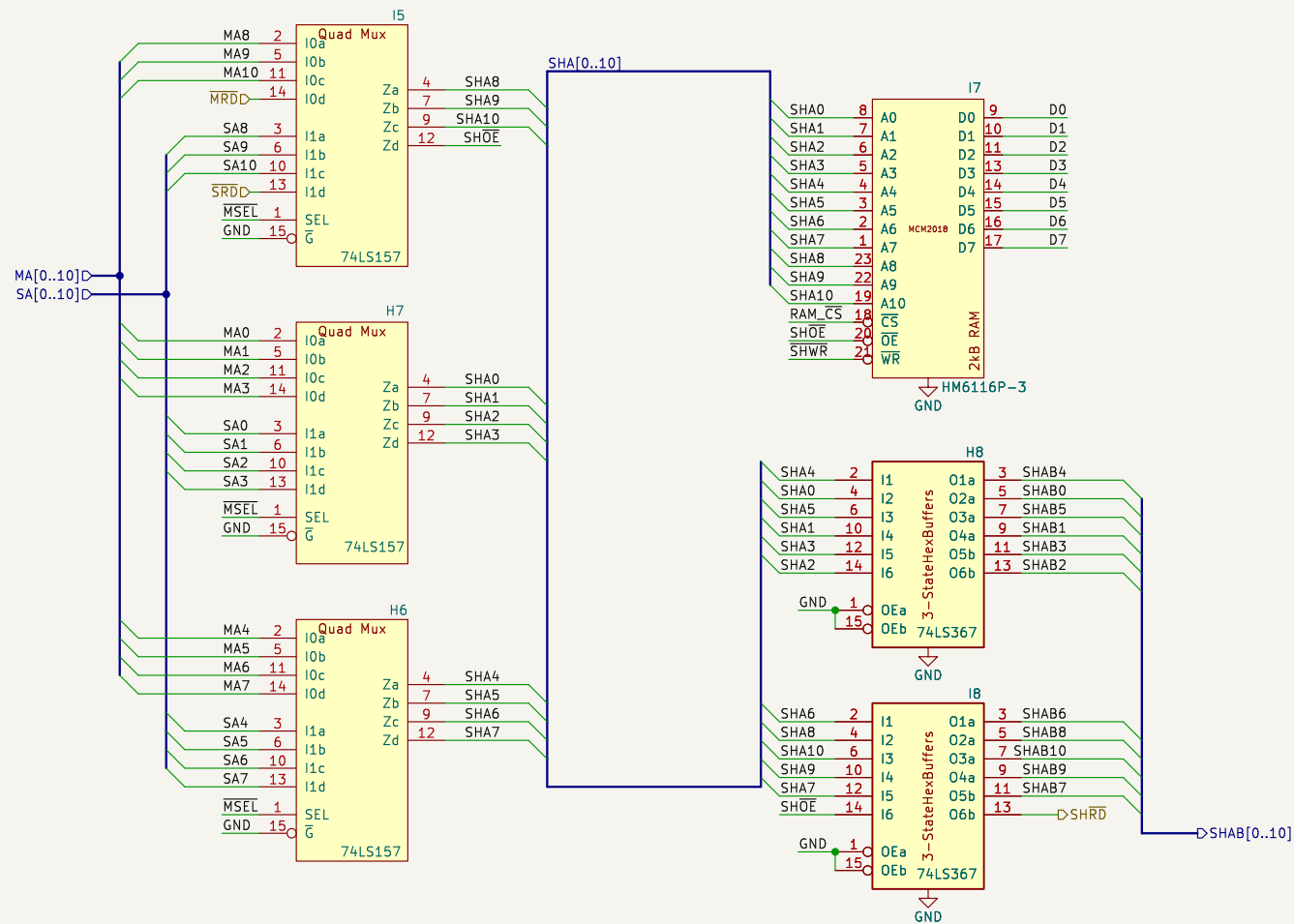
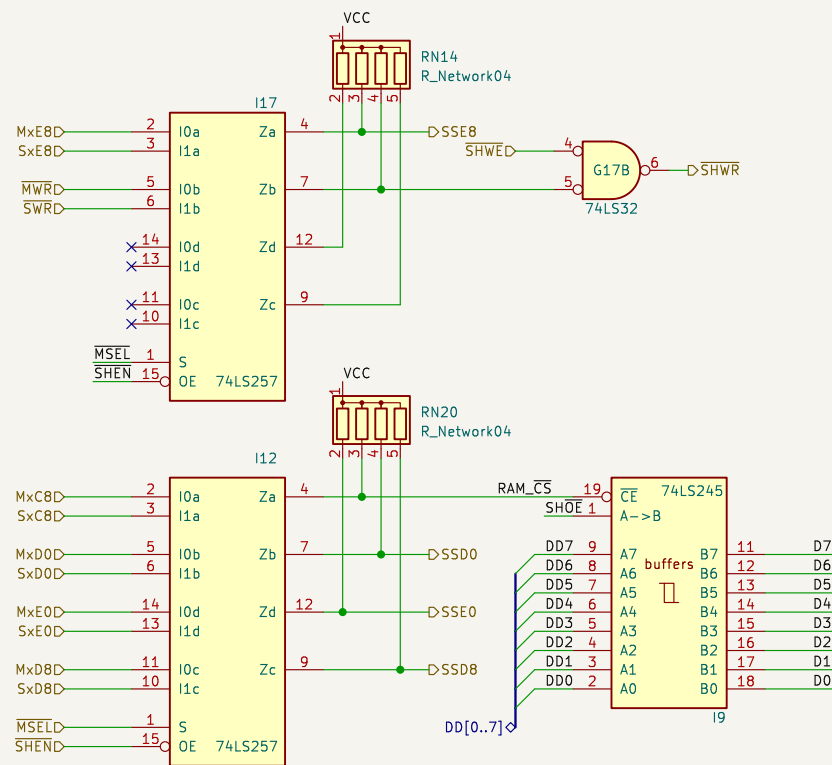
Title: Line Buffer

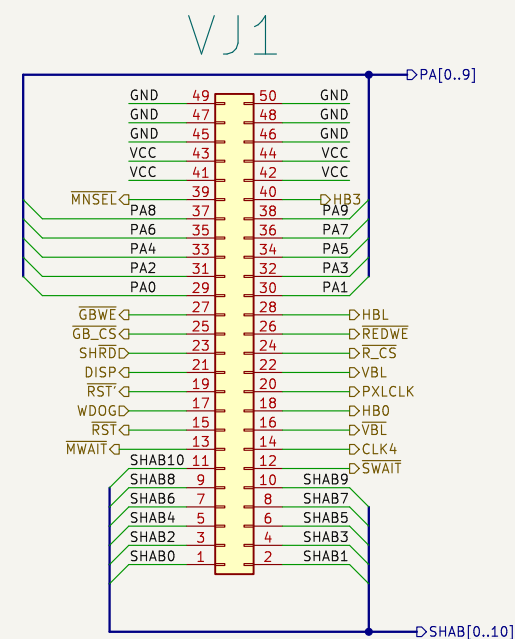
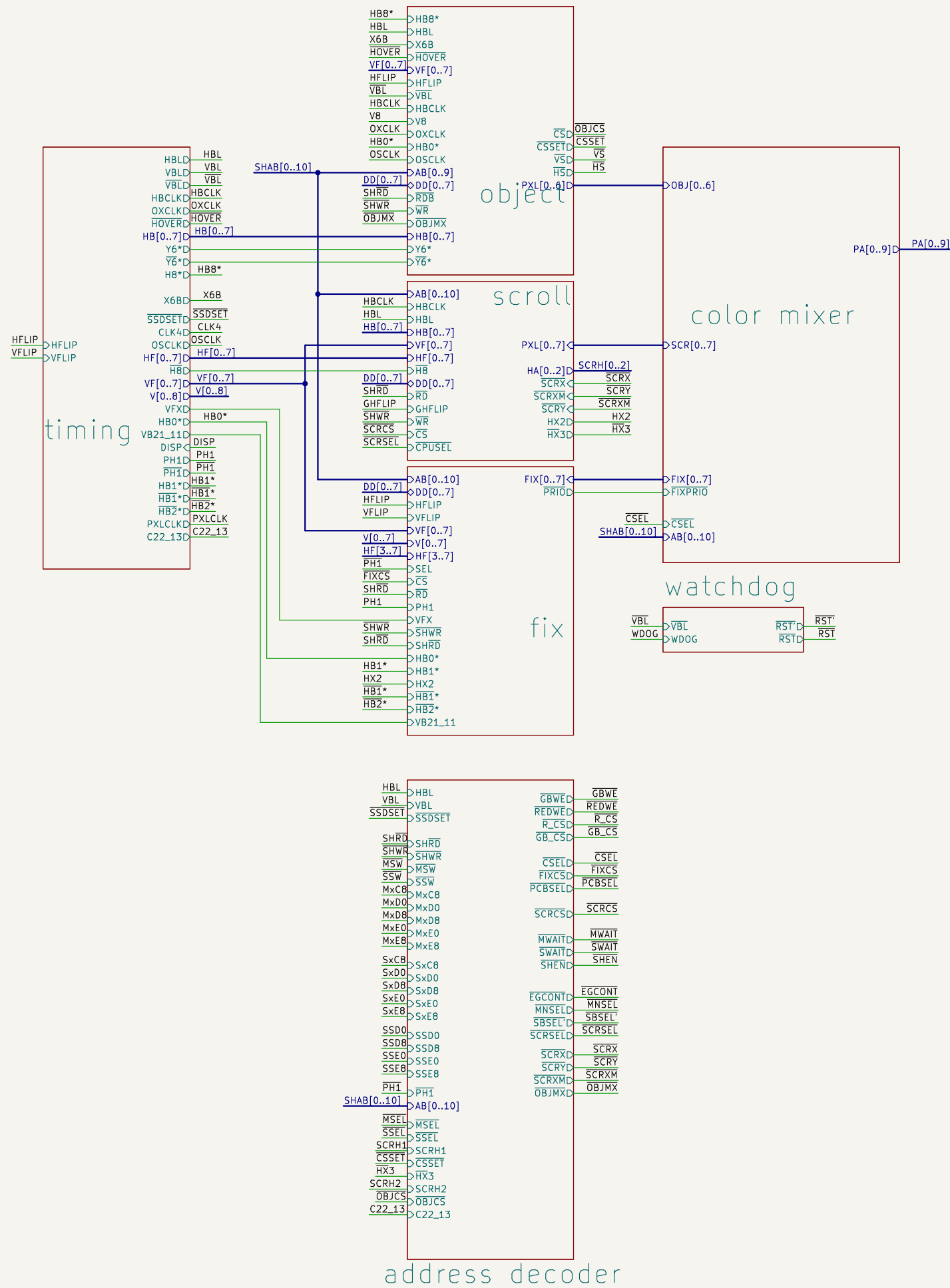
Size: A3
Date: 2024-11-15

Rev:
Id: 2/26



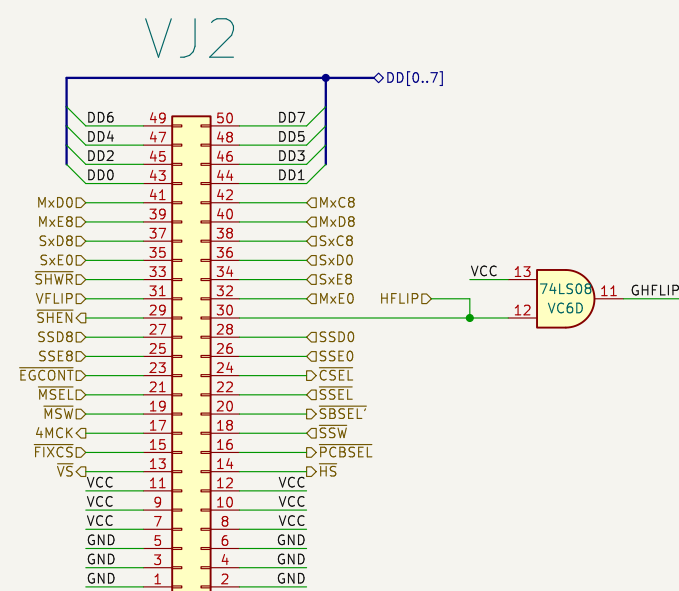






connectors
to CPU board

VJ1 at PCB's top right
VJ2 at PCB's bottom right
pins counted from board bottom up,
pins at the top row of the connector
counted as even



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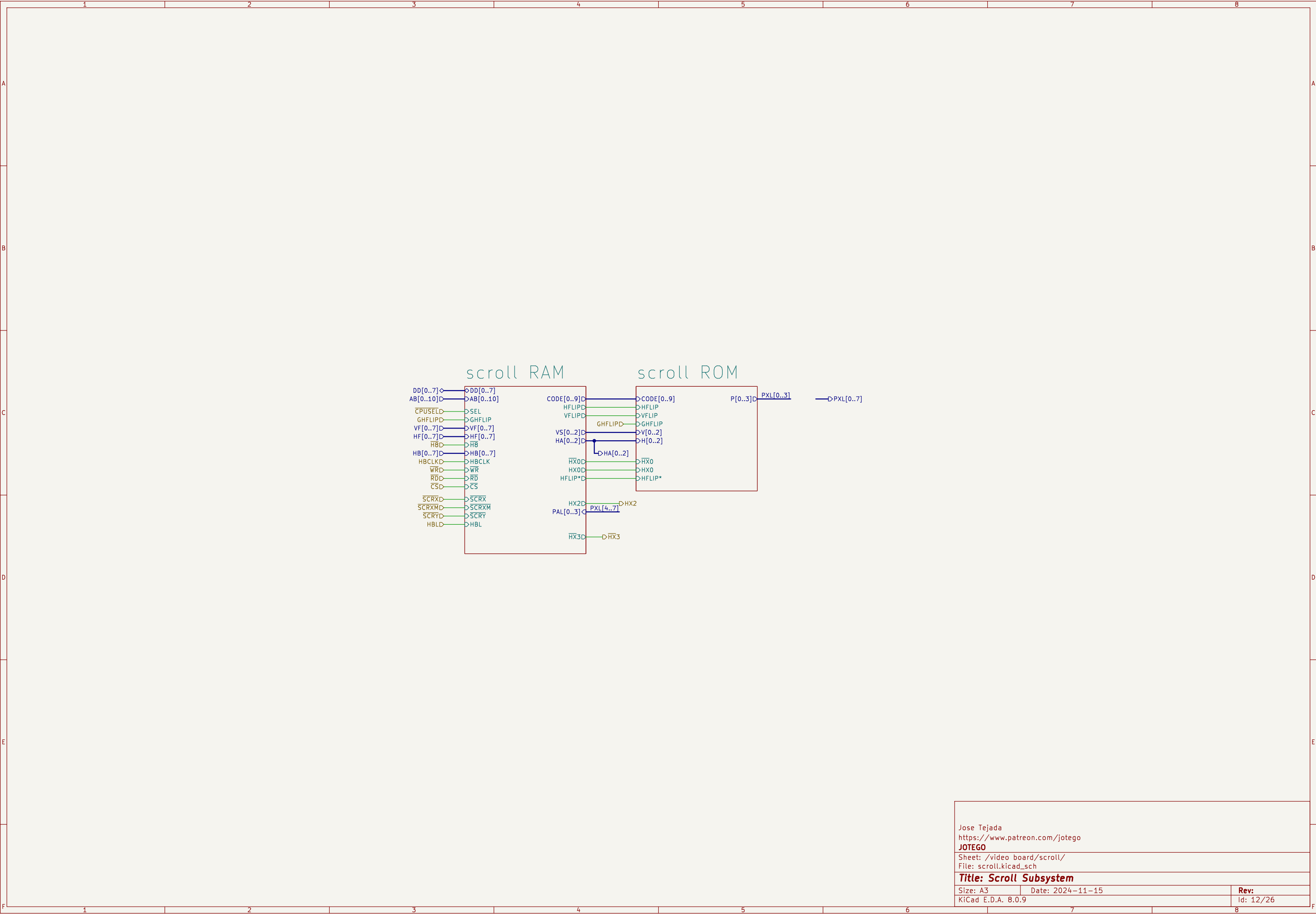
Sheet: /video board/
File: video.kicad_sch

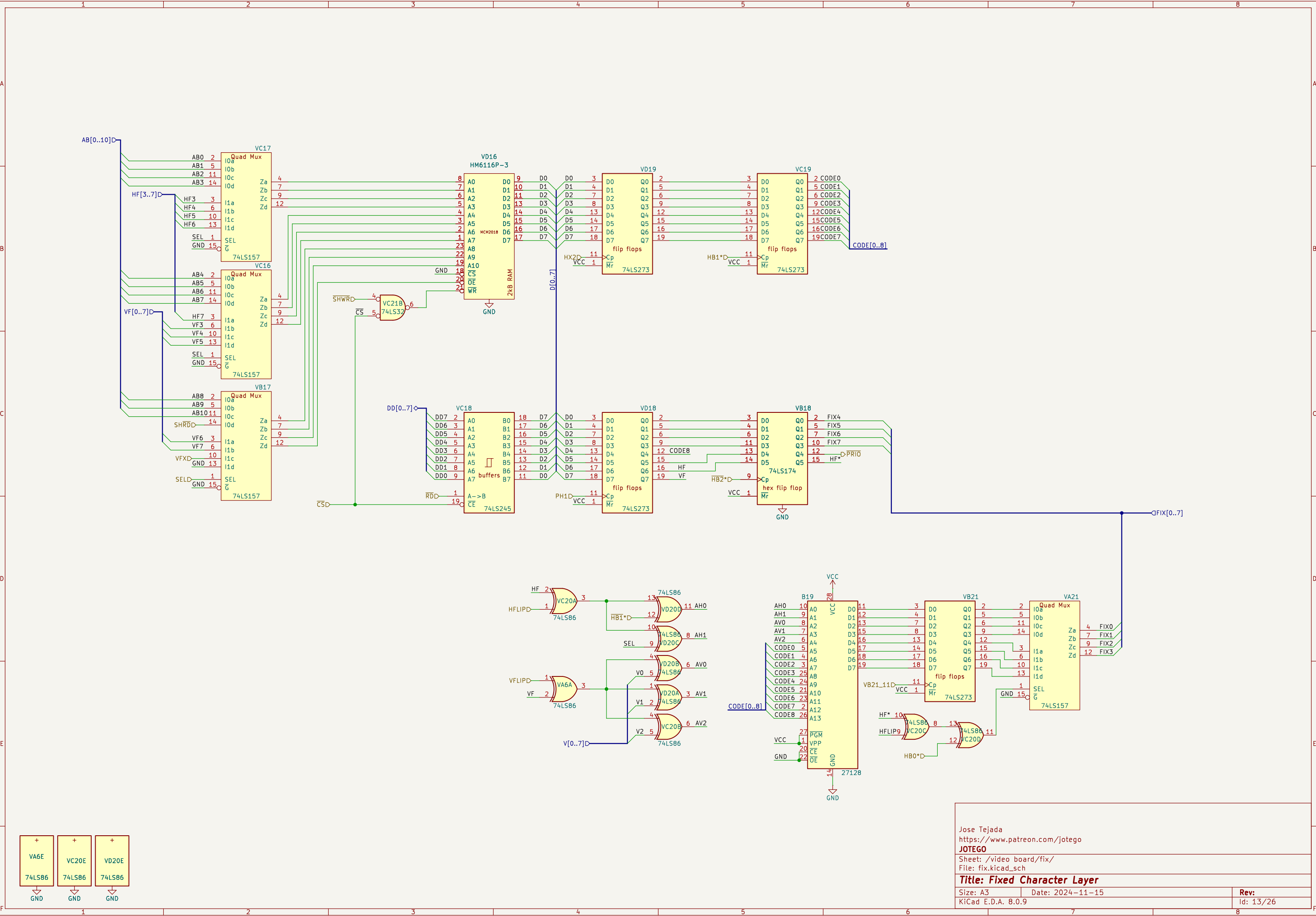
Title: Video board (bottom board)

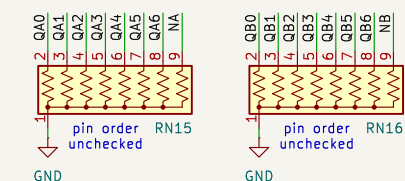
Size: A3
KiCad E.D.A. 8.0.9

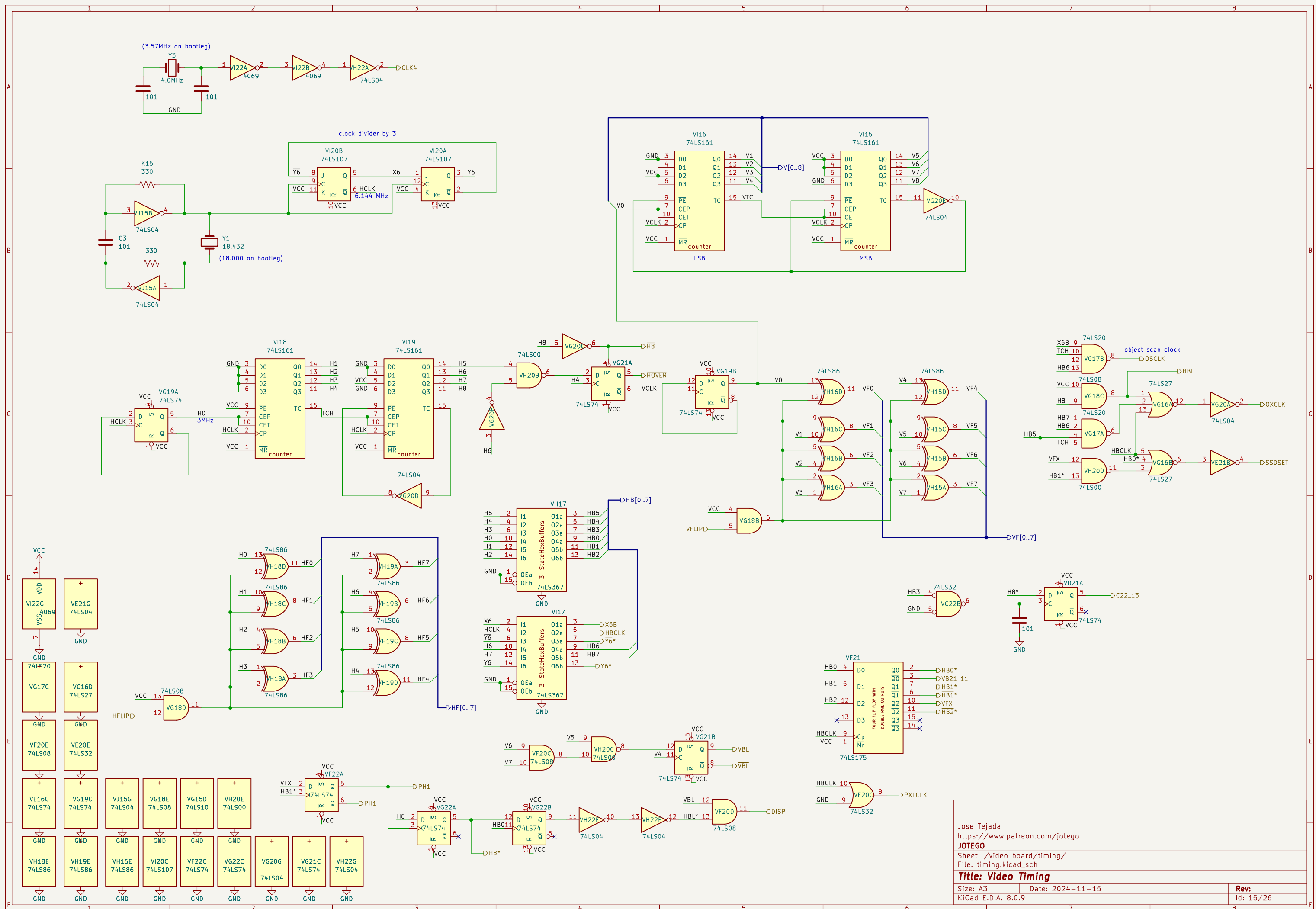
Date: 2024-11-15

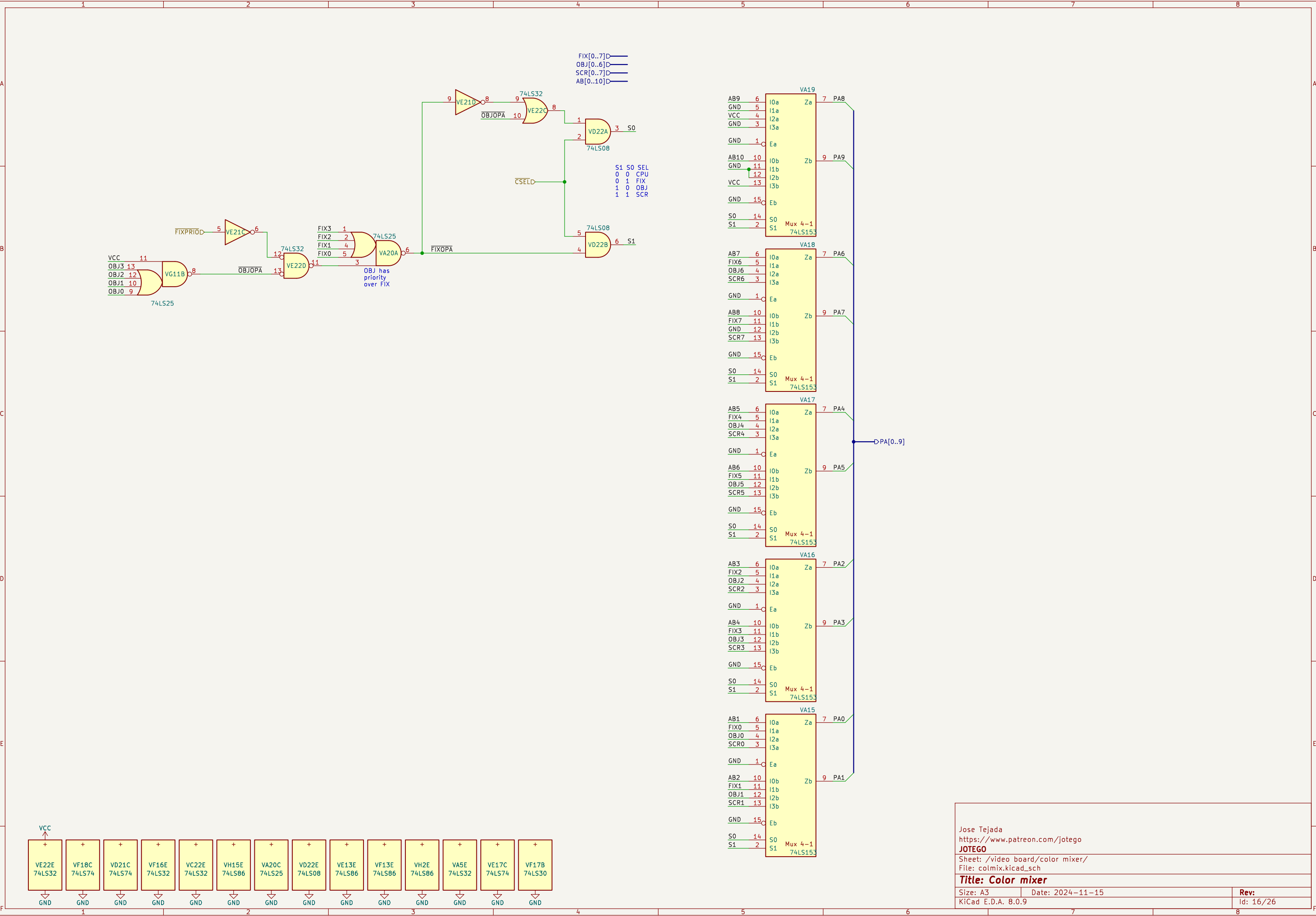
Rev:
Id: 11/26

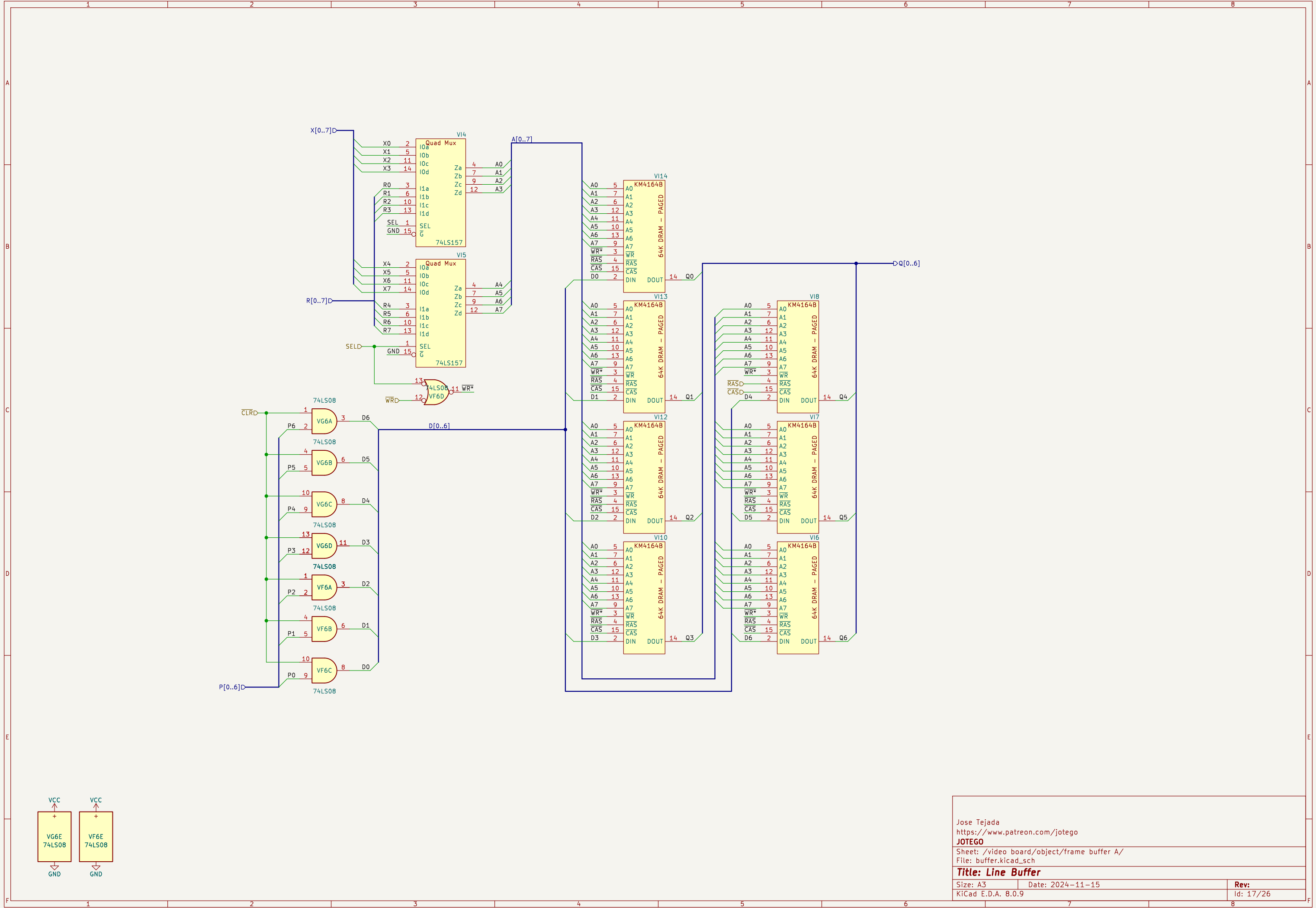


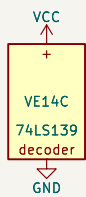




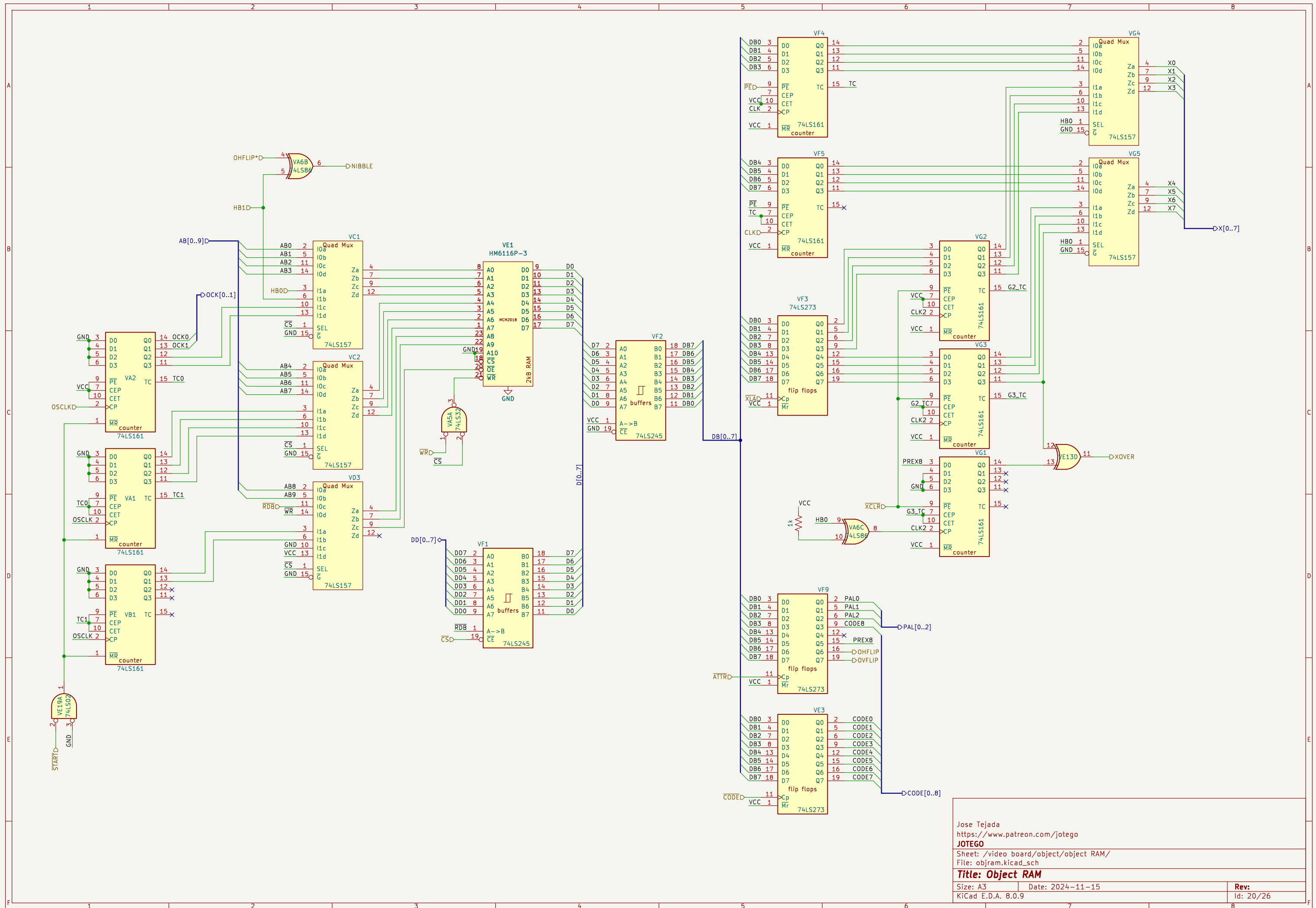








Rev:
Id: 19/26



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Sheet: /video board/object/object RAM/
File: objram.kicad_sch

Title: Object RAM

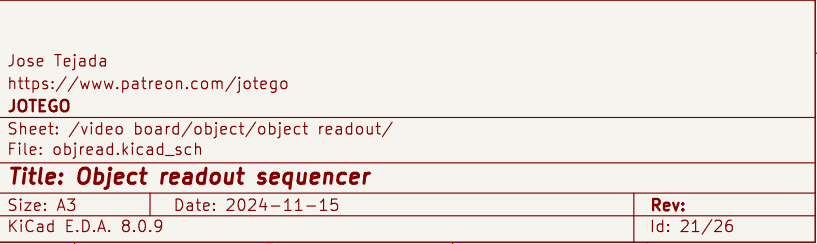
Size: A3

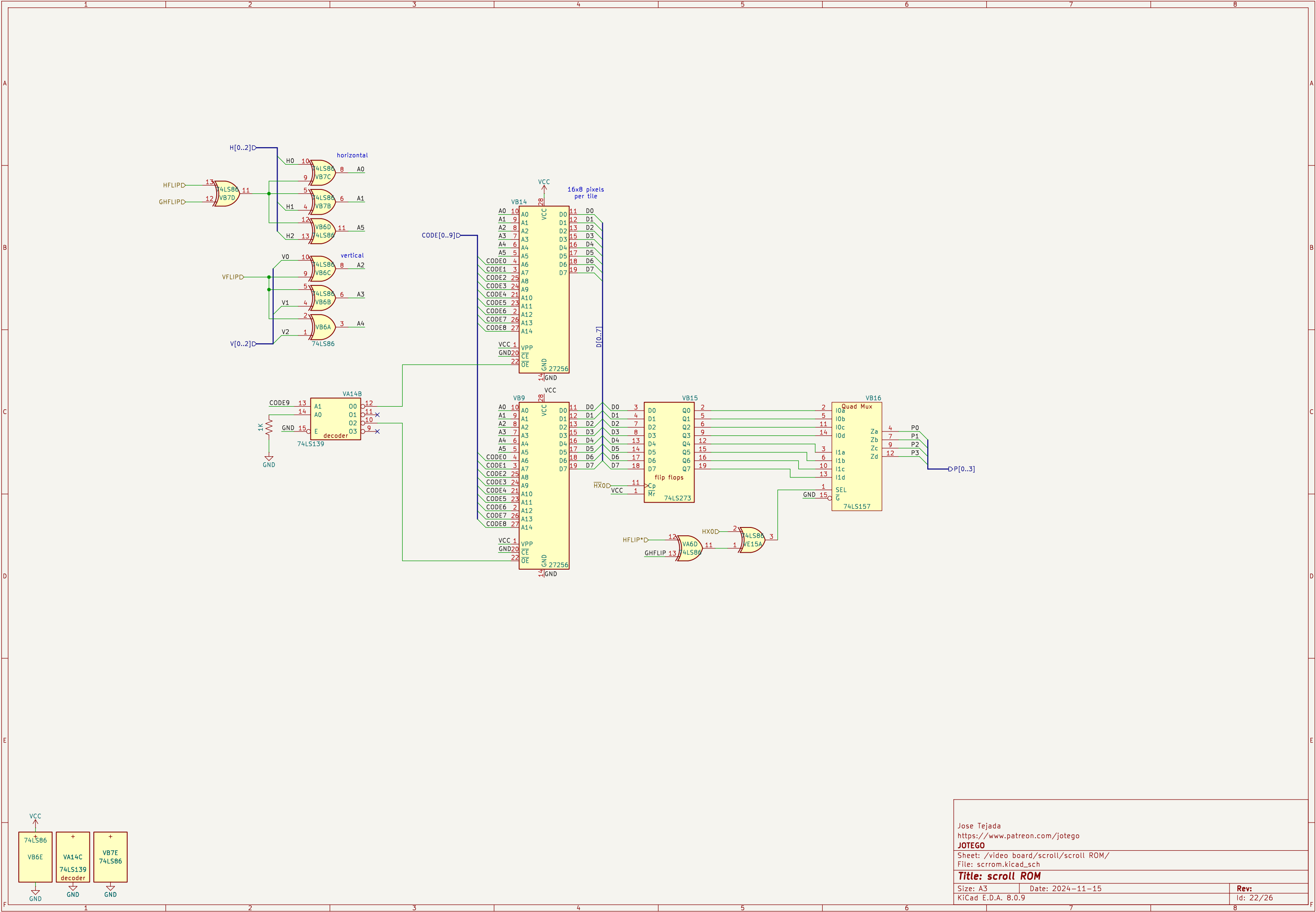
Date: 2024-11-15

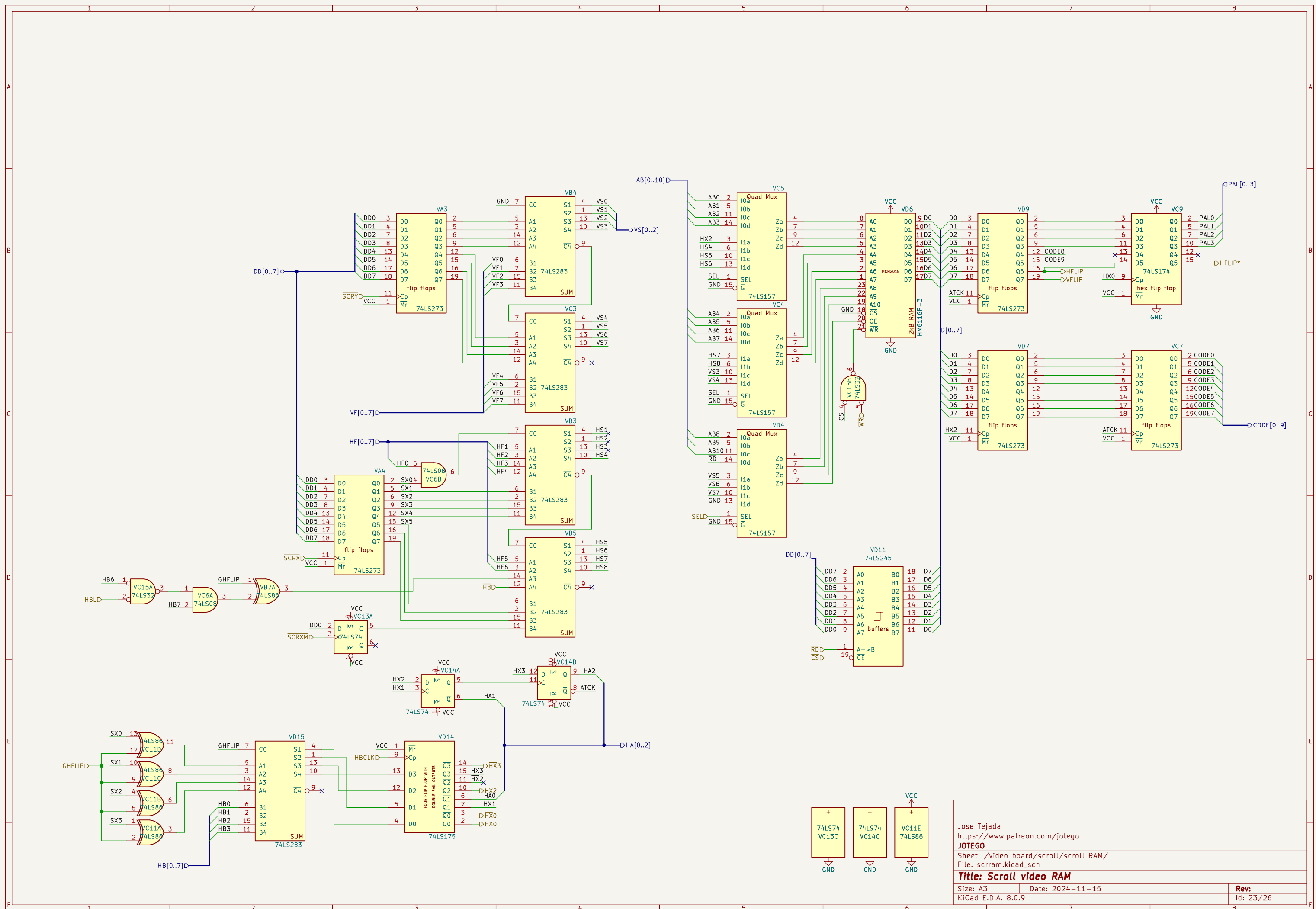
Rev:

KiCad E.D.A. 8.0.9

Id: 20/26

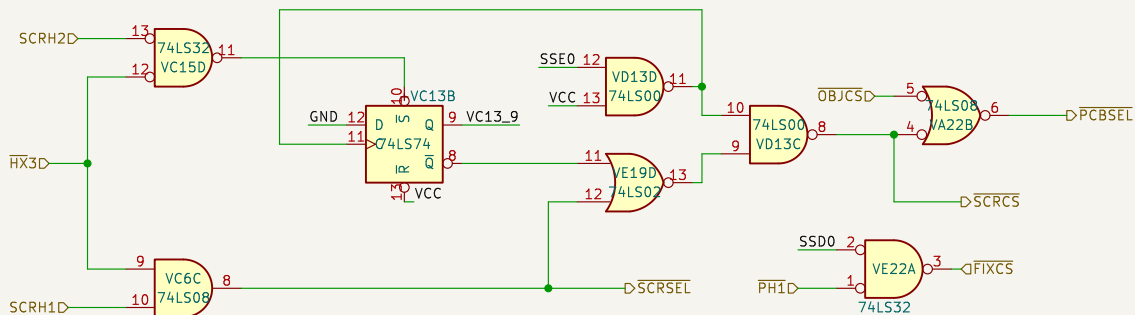


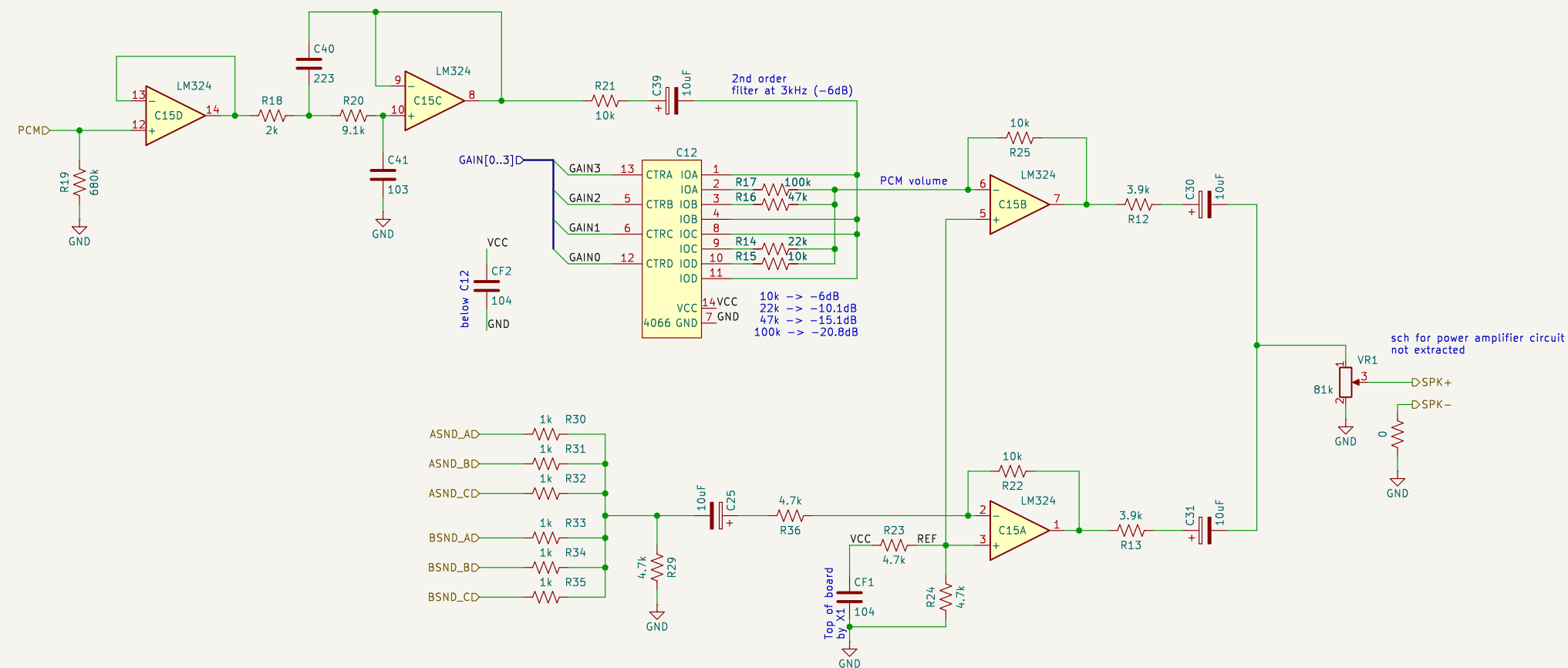






Title: Watchdog Circuit		
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 24/26





Passive component references
when shown, come from original PCB.
The bootleg did not have
references for passives.

Device components follow bootleg references.
The bootleg counts letters from left to right and numbers from bottom to top
The original counts numbers left to right and letters bottom to top.

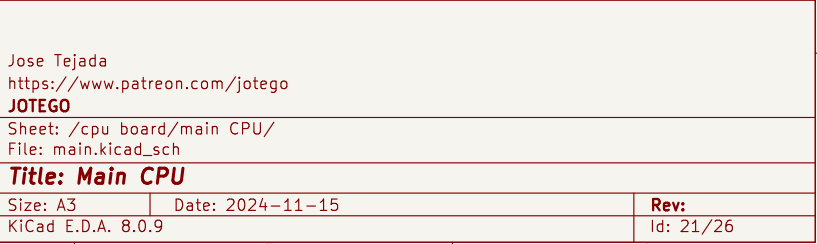
Jose Tejada
<https://www.patreon.com/jotego>
JOTEGO
Sheet: /cpu board/sound/analog/
File: analog.kicad_sch

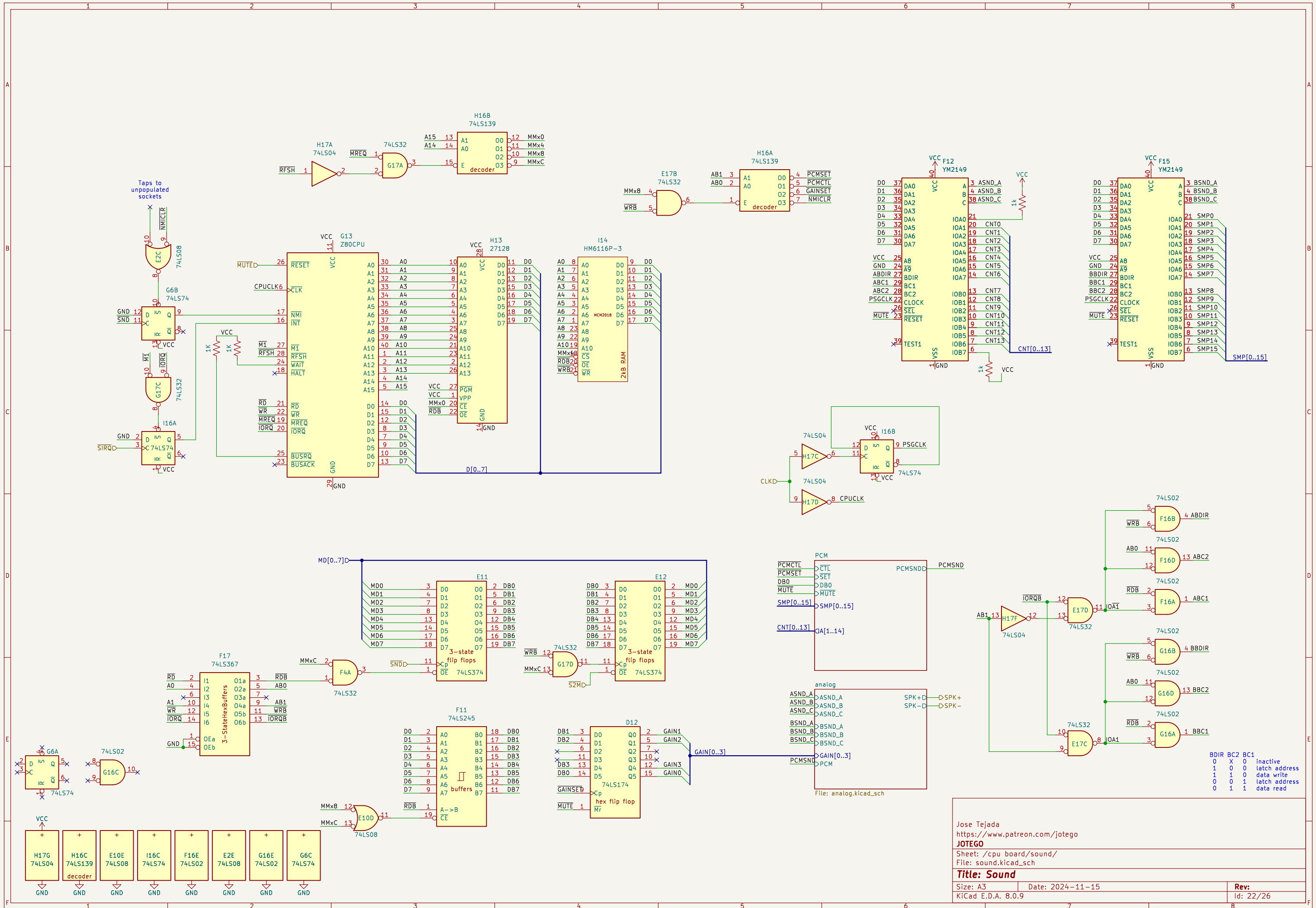
Title: Sound amplifiers

Size: A3 Date: 2024-11-15
KiCad E.D.A. 8.0.9

Rev:
Id: 26/26

VCC
4
V+
C15E
V-
11
LM324
GND



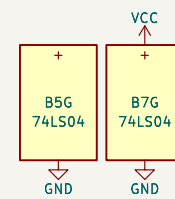
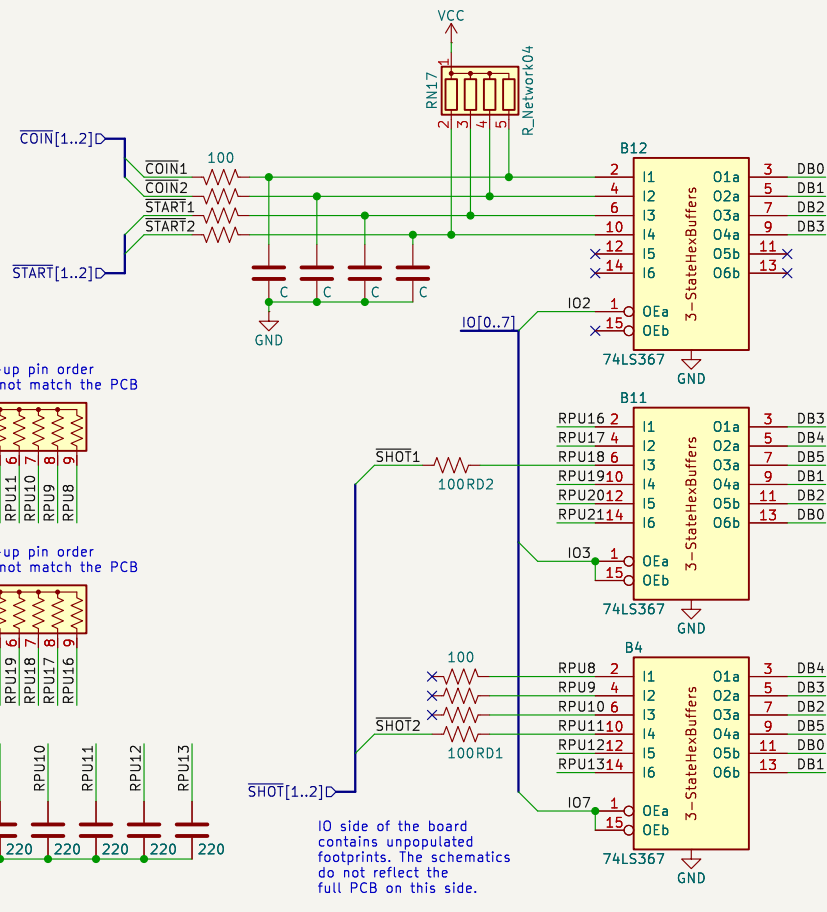




Title: Subordinate CPU

Size: A3	Date: 2024-11-15
KiCad E.D.A. 8.0.9	

Rev:
Id: 23/26



Rev: 24/26

